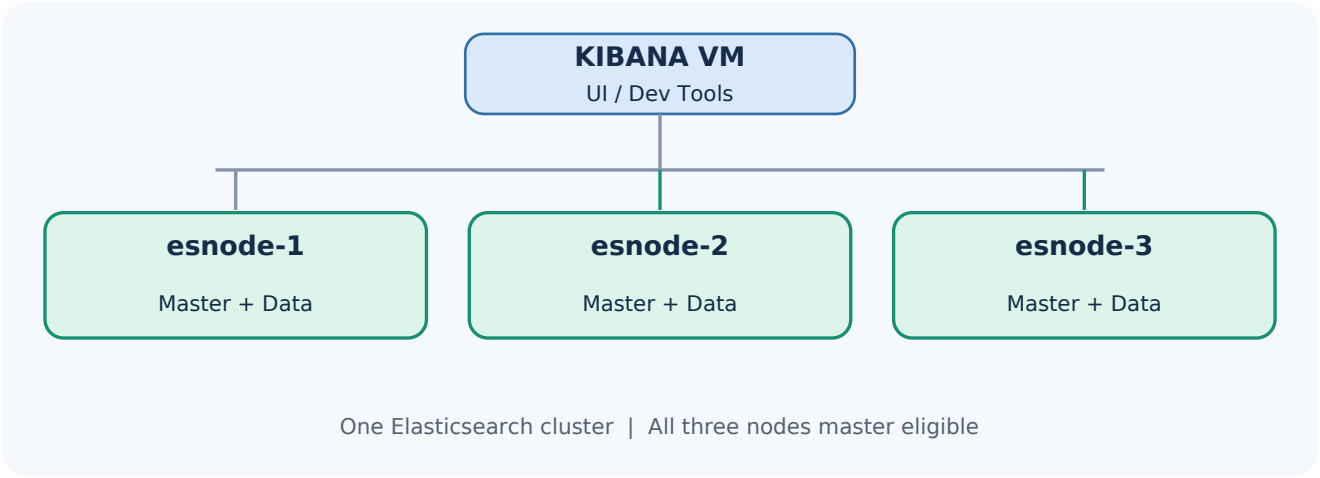
ELK Cluster Node Roles and Hot / Warm Data Tiers

Step-by-step Ubuntu lab manual for the existing 3-node Elasticsearch cluster, separate Kibana VM, and Elastic Observability stack.



Applicable setup: esnode-1, esnode-2, esnode-3; separate Kibana node; Filebeat, Metricbeat, Logstash and Elastic Agent/Fleet integrations.

Training recommendation
Keep all three existing Elasticsearch nodes **master eligible + hot + content + ingest** for the main lab. Use the single-warm-node topology only as a supervised demonstration, or add more VMs for a realistic production-style tier layout.

Learning outcomes

- Identify what master-eligible, data, ingest, coordinating-only and data-tier roles do.
- Inspect current roles, update **node.roles** safely, and verify the three-node cluster.
- Demonstrate data allocation from hot to warm and diagnose allocation failures.

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1 Existing architecture and role concepts

A Kibana server provides the user interface. The Elasticsearch service on each esnode stores/indexes/searches data and can receive or forward client requests. Kibana itself is **not** an Elasticsearch coordinating-only node.

Role / setting	Primary responsibility	In this lab
master	Cluster metadata, index creation and master election	3 eligible voters
data	Generic data: can host any standard tier	Avoid mixing with specialized data_* tiers
data_hot	Recent high-volume logs and metrics	Recommended all 3 initially
data_content	General indices and system indices	Required alongside data_hot unless data is used
data_warm	Older, less-frequently queried time-series shards	Optional demonstration on esnode-3
data_cold / data_frozen	Infrequently accessed / snapshot-based datasets	Not needed for initial training
ingest	Processes Elasticsearch ingest pipelines	Include for observability
transform	Supports Elastic transforms and Fleet-related features	Include for Fleet-related training
remote_cluster_client	Remote clusters and related capabilities	Include where feature requires it
coordinating-only	Forwards requests and merges results; node.roles: []	Only on an additional Elasticsearch instance

Important distinctions

• A **master-eligible** node can become elected master; only one is active at a time. Three master-eligible nodes typically require a majority of two votes to retain quorum.

• Every Elasticsearch node coordinates requests automatically, including data nodes. **node.roles: []** creates a node dedicated to coordination only.

• An unspecified **node.roles** defaults to a broad set of roles, including generic **data**. For physically distinct hot/warm tiers, set explicit roles on each node.

2 Preflight checks before editing

Run all Elasticsearch REST commands below in **Kibana > Dev Tools > Console**. Run the Linux commands in the Ubuntu shell on the named Elasticsearch VM.

Step 2.1 - Record health, node roles, shards and tier settings

```
GET _cluster/health?pretty
GET _cat/nodes?v&h=name,ip,node.role,master
GET _nodes?filter_path=nodes.*.name,nodes.*.roles
GET _cat/allocation?v
GET _cat/shards?v&h=index,shard,prirep,state,node
GET _cluster/settings?include_defaults=false&pretty
```

Expected: three Elasticsearch node names appear. The master column marks the elected master with *. Cluster health should be known before making changes (green preferred; investigate red). Roles output lists the full names, unlike abbreviated **node.role**.

Step 2.2 - Record the current YAML and create a backup on each VM

```
sudo grep -n '^node.roles:' /etc/elasticsearch/elasticsearch.yml
sudo grep -n '^node.name:' /etc/elasticsearch/elasticsearch.yml
sudo cp -a /etc/elasticsearch/elasticsearch.yml /tmp/es-roles-backup.yml
sudo systemctl status elasticsearch --no-pager
```

Step 2.3 - Protect existing data

- Take a verified Elasticsearch snapshot if the indexed data matters. Backing up only **elasticsearch.yml** does **not** back up documents or cluster state.
- Check disk capacity and free space; exclude nodes with unhealthy disks or unresolved unassigned shards from a role migration.
- Keep current cluster.name, node.name, network addresses, discovery and TLS settings. Do not re-bootstrap the established cluster using cluster.initial_master_nodes.

> **Caution: changing roles in an existing cluster**
> Do not remove a data or master role from a populated node without first confirming the on-disk state is compatible and relocating shards where needed. Elasticsearch may refuse to start when repurposed nodes contain unexpected shard or metadata files.

3 Recommended configuration - all three nodes

Use the following design for your current Filebeat/Metricbeat/Logstash/Elastic Agent labs. It preserves three voting master-eligible nodes and places the hot and content tiers on all three machines.

VM	Recommended node.roles
esnode-1	master, data_hot, data_content, ingest, transform, remote_cluster_client
esnode-2	master, data_hot, data_content, ingest, transform, remote_cluster_client
esnode-3	master, data_hot, data_content, ingest, transform, remote_cluster_client
Kibana VM	No Elasticsearch node roles - leave Kibana service separate

Step 3.1 - Edit elasticsearch.yml on esnode-1

```
sudo nano /etc/elasticsearch/elasticsearch.yml
```

Find any existing **node.roles:** line and replace it with the following. Do not add duplicate YAML keys.

```
node.roles: [master, data_hot, data_content, ingest, transform, remote_cluster_client]
```

Step 3.2 - Apply on esnode-2 and esnode-3

Repeat the same edit in each node's local **/etc/elasticsearch/elasticsearch.yml**. Preserve each node's unique **node.name** and network/transport/TLS settings.

Step 3.3 - Restart using a rolling procedure

Apply the changes one node at a time using the detailed procedure in Section 4. A simple **systemctl restart** restarts a service but does not replace the safety and verification steps needed for a live cluster.

Why not set data as well?

The generic **data** role makes a node eligible across all normal data tiers and takes precedence over specialized tier roles. For an explicit tier layout, use **data_hot**, **data_content** and, where appropriate, **data_warm** without **data**.

Expected behavior
The recommended roles provide a stable training cluster but do not create a physically distinct warm tier. Hot and warm relocation can only be demonstrated after at least one warm-role node is available.

4 Controlled rolling restart - one node at a time

The sequence below follows the Elastic guidance for self-managed rolling restarts. Perform it during a maintenance window. Repeat separately for esnode-1, then esnode-2, then esnode-3.

Step 4.1 - Optional: pause nonessential ingestion

For a training lab, pause noisy writers/ingest jobs when practical, or allow them to continue if their availability is important.

Step 4.2 - Temporarily permit primary allocation only

```
PUT _cluster/settings
{
  "persistent": {
    "cluster.routing.allocation.enable": "primaries"
  }
}
```

Step 4.3 - Stop and start the selected Elasticsearch VM

```
sudo systemctl stop elasticsearch
# Modify node.roles in elasticsearch.yml if not yet done.
sudo systemctl start elasticsearch
sudo systemctl status elasticsearch --no-pager
sudo journalctl -u elasticsearch -n 60 --no-pager
```

Step 4.4 - Verify rejoin, restore shard allocation

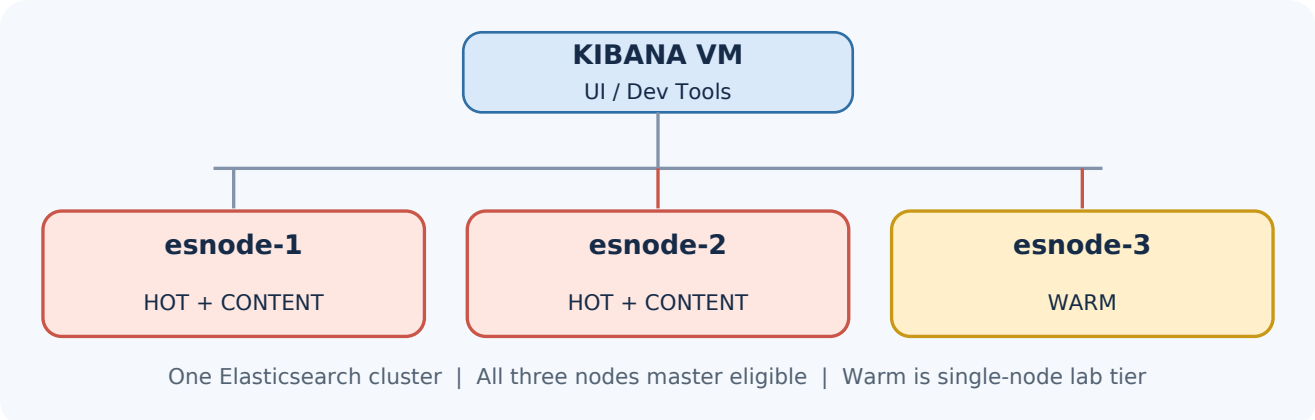
```
GET _cat/nodes?v&h=name,ip,node.role,master
GET _cluster/health?pretty
PUT _cluster/settings
{
  "persistent": {
    "cluster.routing.allocation.enable": null
  }
}
GET _cat/recovery?v
```

Wait for the restarted node to rejoin and cluster recovery to stabilize before updating the next node. An intermediate **yellow** status can occur while replicas are unavailable; investigate any persistent **red** status.

> **Do not skip recovery**
> With three master-eligible nodes, stop only one at a time. Never leave replica allocation disabled after maintenance, and do not proceed to the next node while the cluster remains unexpectedly unhealthy.

5 Alternative lab - dedicated hot and warm tiers

Use this **only if you specifically need to demonstrate physical warm-tier allocation** and the current shards have been assessed. A safer extension is to add a separate warm VM instead of converting esnode-3.



Node	Explicit role setting
esnode-1 and esnode-2	master, data_hot, data_content, ingest, transform, remote_cluster_client
esnode-3 (warm)	master, data_warm

Step 5.1 - Inspect shards and existing tier preferences

```
GET _cat/shards?v&h=index,shard,prirep,state,node
GET _cat/allocation?v
GET _all/_settings/index.routing.allocation.*?flat_settings
```

Step 5.2 - Prepare and drain esnode-3 appropriately

Review each shard on esnode-3, including system/content shards. Confirm esnode-1 and esnode-2 can accept the relevant hot/content data, disk usage permits relocation, and index allocation filters/ILM policies allow it. Move shards with supported allocation settings, then check relocation has completed. Only then remove the hot/content roles and restart esnode-3 according to the role-change procedure.

Single warm node = lab limitation
Warm indices configured with one replica typically remain **yellow** because a primary and replica cannot be on the same node. A production-style redundant warm tier needs at least two warm-eligible nodes. Keeping esnode-3 master eligible also adds control-plane workload to the warm VM.

6 Practical hot-to-warm shard allocation exercise

Prerequisite: two nodes already provide data_hot and one node provides data_warm, and your cluster is stable. These commands create a **new disposable test index**; do not apply the preference change to production indices.

Step 6.1 - Create an index preferring hot

```
PUT lab-tier-demo-001
{
  "settings": {
    "number_of_shards": 1,
    "number_of_replicas": 0,
    "index.routing.allocation.include._tier_preference": "data_hot"
  }
}
```

Step 6.2 - Send a sample document

```
POST lab-tier-demo-001/_doc
{
  "message": "Hot tier demonstration",
  "level": "INFO"
}
```

Step 6.3 - Observe where Elasticsearch stores it

```
GET _cat/shards/lab-tier-demo-001?v&h=index,shard,prirep,state,node
```

Expected example: STARTED primary shard on esnode-1 or esnode-2. Exact node and shard IDs can vary.

Step 6.4 - Change preference to warm, with hot fallback

```
PUT lab-tier-demo-001/_settings
{
  "index.routing.allocation.include._tier_preference": "data_warm,data_hot"
}
```

Step 6.5 - Verify the result

```
GET _cat/shards/lab-tier-demo-001?v&h=index,shard,prirep,state,node
GET lab-tier-demo-001/_settings?filter_path=*.settings.index.routing.allocation.include._tier_preference
```

Wait for relocation to complete. If esnode-3 is the only eligible warm node and has capacity, the primary should appear on esnode-3. The test uses zero replicas deliberately to demonstrate placement with one warm VM.

7 ILM automation for a real hot-to-warm lifecycle

In an operational deployment, use **Index Lifecycle Management (ILM)** to transition older indices automatically. This exercise describes the configuration approach; it does not modify your existing Metricbeat/Filebeat/Fleet data streams.

Step 7.1 - Create a lab ILM policy in Kibana

- Go to **Stack Management > Index Lifecycle Policies** (location may vary with Elastic version).
- Create a policy such as **lab-hot-warm-policy**. Configure the hot phase for active data.
- Enable a warm phase after a training threshold, such as **1 day**. An ILM migrate action normally updates the index tier preference automatically.
- Optionally add a delete phase for disposable lab data only. Do not set a delete policy for real data without retention approval.

Step 7.2 - Attach the policy to a lab index template

Under **Stack Management > Index Management > Index Templates**, create a template matching a new lab data stream/index pattern. Assign the lifecycle policy and any required rollover settings. Avoid overriding Elastic-managed integration templates.

Step 7.3 - Verify phases and tier preference

```
GET lab-tier-demo-001/_ilm/explain
GET lab-tier-demo-001/_settings?filter_path=*.settings.index.lifecycle*,*.settings.index.routing.allocation.include._tier
GET _cat/shards/lab-tier-demo-001?v&h=index,shard,state,node
```

Note: The manually created Section 6 index will not enter your new ILM policy automatically. Associate a policy to an index/template first and meet the lifecycle conditions.

How the stages relate to observability

Stage	Typical data	Expected location
HOT	Fresh Filebeat logs; Metricbeat and Elastic Agent metrics	Fast hot nodes
WARM	Older time-series indices, read infrequently	Warm nodes
COLD / FROZEN	Historical/archive data where supported	Dedicated cold/frozen configurations

8 Coordinating-only node and common problems

Optional: add a fourth Elasticsearch node for coordination

If you want to demonstrate a dedicated coordinating-only role, run a **separate fourth Elasticsearch instance**, preferably on a new VM, that joins the existing cluster. It will not hold shard data or participate in master elections. The Kibana application alone does not have this role.

```
# In the NEW Elasticsearch node's elasticsearch.yml:
node.roles: []
# Supply the existing cluster.name, discovery and security/TLS settings.
# Do not remove roles from any existing data/master VM just to show this.
```

Troubleshooting quick reference

Symptom	Diagnostic / correction
Node will not start	sudo journalctl -u elasticsearch -n 120 --no-pager; check malformed YAML, role repurposing restrictions and TLS.
Missing esnode-3	GET _cat/nodes?v; check service, transport connectivity, config and logs.
Cluster RED	GET _cluster/health; GET _cat/shards; resolve unassigned primaries before migration.
Cluster YELLOW after warm move	A lone warm node cannot host primary plus replica. Add warm capacity or change replica count only on disposable lab data.
Shard stuck HOT	GET _cluster/allocation/explain; check available warm node, disk watermarks and conflicting filters.
ILM is not migrating	GET your-index/_ilm/explain; verify policy binding, phase age, rollover and warm tier existence.

Useful verification commands

```
GET _cluster/health?pretty
GET _cat/nodes?v&h=name,ip,node.role,master
GET _nodes?filter_path=nodes.*.name,nodes.*.roles
GET _cat/allocation?v
GET _cat/shards?v&h=index,shard,prirep,state,node
GET _ilm/status
```

9 Lab completion checklist and expected evidence

Check	Expected observation
Elasticsearch service	Active (running) on each of the three esnodes
Node enumeration	Three Elasticsearch nodes visible in _cat/nodes
Master election	Exactly one elected master (*) with the others eligible
Recommended role mapping	All three nodes show master, data_hot, data_content, ingest, transform and remote_cluster_client
Basic observability	Filebeat / Metricbeat / Elastic Agent data continues to index and appear in Kibana Discover
Optional warm lab	esnode-3 has data_warm; a test primary shard moves to esnode-3
Replica/cluster health	No unexpected unassigned primaries; known single-warm replica limitation acknowledged
ILM exercise	Lifecycle policy, phase transitions and tier preference observable via API

Post-lab cleanup

```
# Optional: delete ONLY the disposable lab test index
DELETE lab-tier-demo-001

# Always make sure shard allocation was restored:
GET _cluster/settings?include_defaults=false&pretty
```

If the temporary **cluster.routing.allocation.enable: primaries** setting is still present after maintenance, reset it to **null** using Section 4. Remove temporary training policies/templates when they are no longer required.

Official Elastic references

Node settings: https://www.elastic.co/docs/reference/elasticsearch/configuration-reference/node-settings

Node roles: https://www.elastic.co/docs/deploy-manage/distributed-architecture/clusters-nodes-shards/node-roles

Self-managed data tiers:
https://www.elastic.co/docs/manage-data/lifecycle/data-tiers/manage-data-tiers-self-managed-eck

Tier allocation setting: https://www.elastic.co/docs/reference/elasticsearch/index-settings/data-tier-allocation

Rolling restart procedure:
https://www.elastic.co/docs/deploy-manage/maintenance/start-stop-services/full-cluster-restart-rolling-restart-procedures

Data tiers overview: https://www.elastic.co/docs/manage-data/lifecycle/data-tiers

Design choice: For the current three-node ELK training environment, keeping three combined master/hot/content nodes is the lowest-risk baseline. Dedicated warm and coordinating VMs are better treated as expansion exercises.